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BOLD & BRAVE

- 2023 space policy opens doors for private players
- Lithium reserves in J&K, Rajasthan to boost EV push
- ISRO's moon mission, Chandrayaan-3, to launch in July

BUT...

<2%
India's share in the space economy. Why? 75% of private firms make low-value products

Reaching for the Stars

The ministry has pushed collaborations, be it private sector in space, PSUs for N-power stations or scientific research

By Amarnath K. Menon

WITH THE 2023 SPACE POLICY, the government cleared the decks for India to take a quantum jump into the Second Space Age. For one, the Indian Space Research Organisation (ISRO) will transition out of manufacturing of operational space systems and lean more on non-government entities (NGEs). The private sector thus becomes a critical stakeholder in the space economy.

ISRO is the sixth largest space agency in the world and also ranks fifth globally in the private space tech sector, with over 400 companies operating. But around 75 per cent are involved in manufacturing products in the low-to-medium value add category, one reason why India's share in the global space economy is less than two per cent. ISRO is now focusing on Chandrayaan (moon mission) and Gaganyaan (human spaceflight mission). The first human spaceflight is scheduled for the last quarter of 2024.

The ministry's investment in the science and technology (S&T) system more than doubled in the past nine years to Rs 6,002 crore in 2022-23. Now India ranks third, after the US and China, in the number of PhDs awarded in science and engineering (nearly 25,000). It is also the third most attractive investment destination for technology transactions. The National Initiative for Developing and Harnessing Innovations (NIDHI), launched in 2019, has led to India leapfrogging to No. 40 in the Global Innovation Index (GII).

The ministry is also excited at the discovery of lithium reserves—a key component for batteries of electric vehicles (EVs)—in J&K and Rajasthan. Officials claim the finds could service up to 80 per cent of the country's lithium requirements, which is good news for the government's plan to increase EV penetration to 30 per cent by 2030.

Meanwhile, the department of atomic energy (DAE) has got the go-ahead on joint ventures with PSUs for resources to set up N-power stations. It has already given bulk approval for installing of 10 nuclear reactors. And the DAE's PPP-based research reactor project, envisaged in 2021 for the production of a range of nuclear medicines, could transform India's stature in the healthcare industry. It should be online in five years. ■